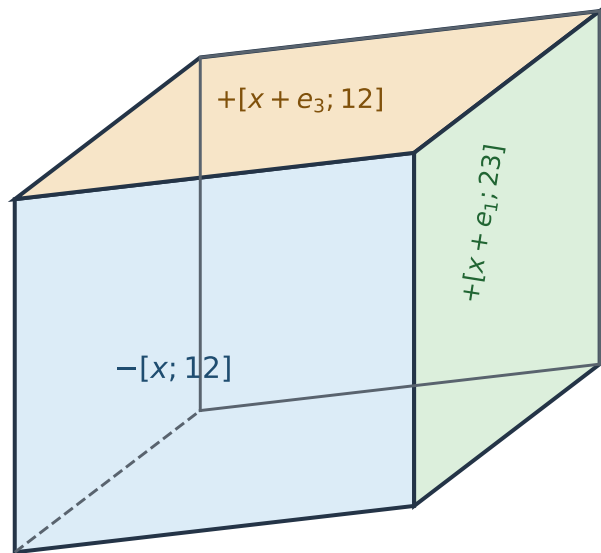


Cellular origin of the carrier



one oriented cube boundary

C_3



∂_3

C_2



∂_2

C_1

$$\partial_2 \partial_3 = 0$$

cube boundaries

$$\dim \operatorname{im} \partial_3 = L^3 - 1$$

wrapping classes

$$\dim H_2(T^3) = 3$$

$$\ker \partial_2 = \operatorname{im} \partial_3 \oplus H_2, \quad \dim \ker \partial_2 = L^3 + 2$$